

Exercise 04: *cocotb* Test

Objective: *Introduction to the cocotb tests.*

Task: *Development of a cocotb test using the Saturation Filter.*

In this exercise you need to develop a simple *cocotb* testbench for a saturation filter (SAT Filter). First make yourself familiar with the DUT. Do not look at the RTL, but rather read the provided documentation in <ROOT>/lectures/common/docs/exercise_general_info.pdf.

Part 1: Simple Test

<ROOT>/lecture03/exercises/E04_sat_cocotb_test/tb contains a **Makefile** which already correctly imports the SAT filter RTL from <ROOT>/lecture03/exercises/E04_sat_cocotb_test/rtl. You only need to update **COCOTB_TEST_MODULES** with you test module name.

Create a new *cocotb* test with a single testcase, which:

- Starts a clock on **clk**.
- Resets the DUT using **rst**.
- Sends two transaction through the DUT.
 - A transaction with a value below the **THRESHOLD**.
 - A transaction with a value above the **THRESHOLD**.
 - Be sure to adhere to the *sSDT* Protocol.
- Check the received outputs using the **assert** statement.

Implement your test in the file: **sat_default_test.py** and add it to the **Makefiles** **COCOTB_TEST_MODULES** variable.

Run the test using the **Makefile** and inspect the waveforms.

Part 2: Multiple transactions

Extend the testcase to drive multiple transactions.

- Generate 10 sets of random input data using python's **random** module;
- Loop over the generated input data. In each iteration:
 - Send the transaction though the DUT.
 - Check the received output using the **assert** statement.

Observe the test results and the waveforms.